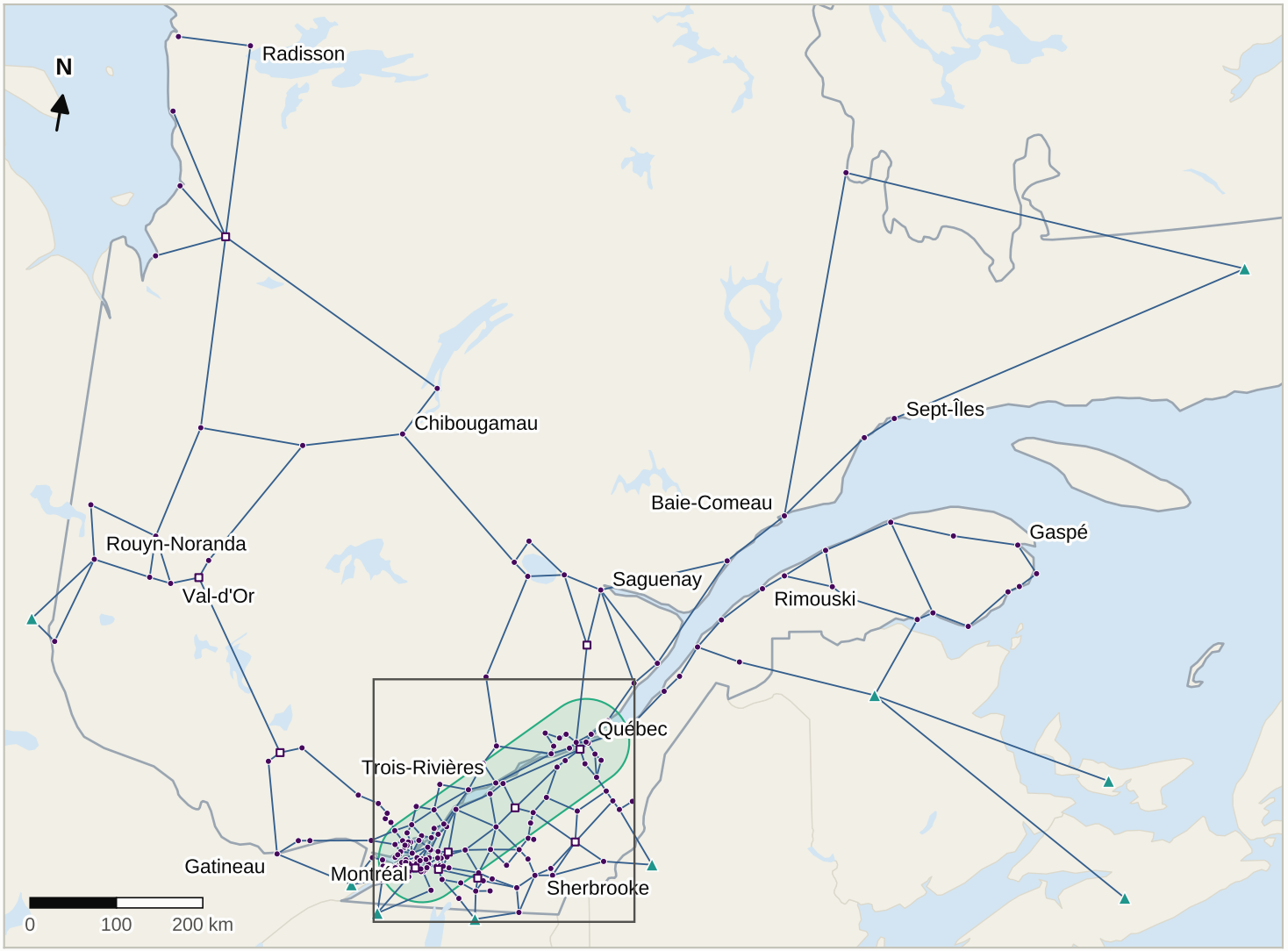
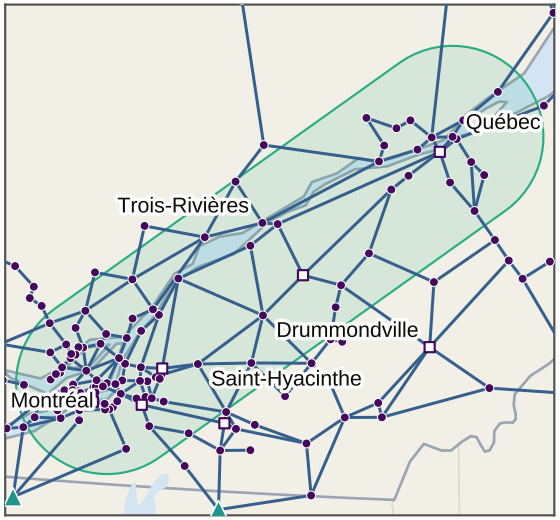
99 of 187 linked cities (53%) lie within 50 km of the Montréal–Québec axis

Cities (nodes) joined by direct inter-city road links, drawn as straight lines · 20 unlinked communities, mostly in Nunavik, not shown



Zoom: Montréal–Québec corridor



Network elements

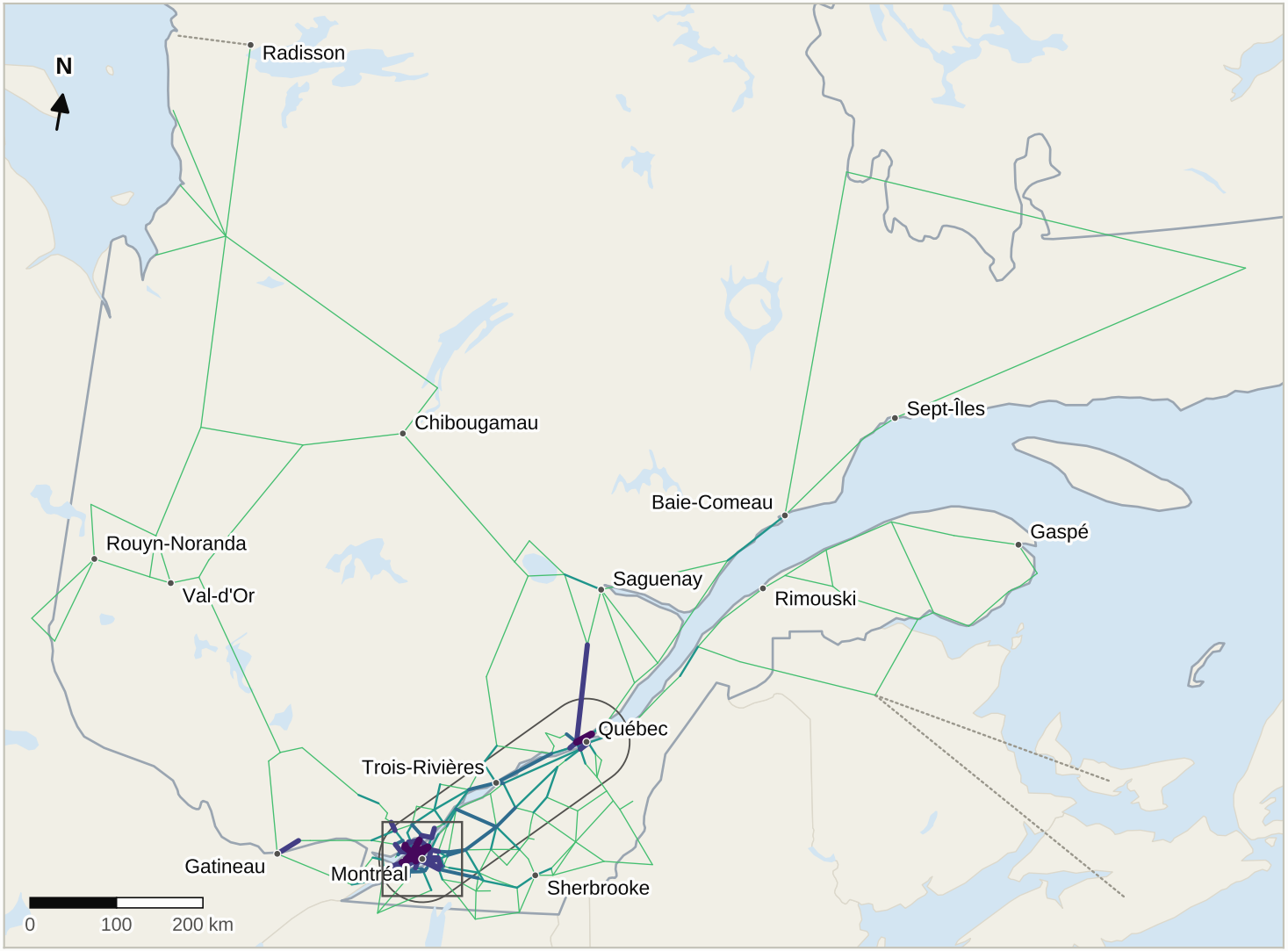
- City (187)
- Junction node (11)
- Border gateway (9)
- Road link (307)
- Corridor: 50 km either side of the Montréal–Québec axis

Network: CIRANO road graph · Basemap: Natural Earth. Analysis: A. Bouznad, CIRANO / Polytechnique Montréal

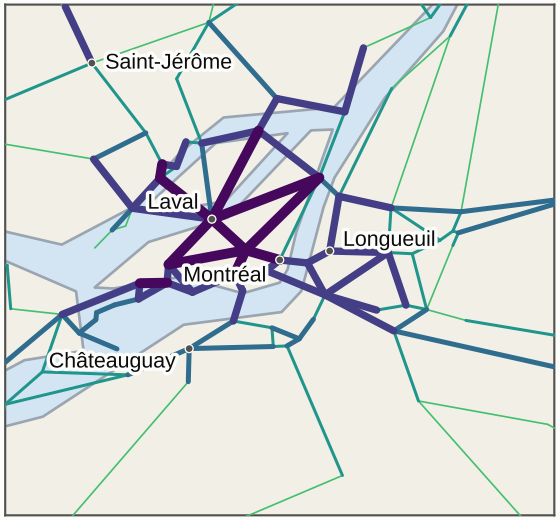
Question	Where are the cities and links of the inter-city road graph, and how concentrated is the network?
Data & method	207 municipalities (187 linked), 11 junction nodes and 307 links built for the CIRANO resilience project; corridor = 50 km either side of the straight Montréal–Québec axis.
What it helps decide	Sets the reading frame for every result that follows: corridor and remote-region links must be compared within their own context, not on one national scale.

Montréal–Québec corridor links carry 65% of vehicle-km on 28% of network length

Annual average daily traffic (AADT) per link, 2025, vehicles/day, both directions · 5 natural-breaks (Jenks) classes, line width and colour



Zoom: Greater Montréal



AADT, vehicles/day (links)

- 112,286 – 185,912 (12)
- 64,000 – 102,842 (40)
- 36,900 – 60,445 (32)
- 16,000 – 35,000 (68)
- 530 – 15,513 (152)
- No MTMD data (3)
- Montréal–Québec corridor (50 km either side of the axis)

Data: MTMD traffic counts (AADT), Données Québec, CC BY 4.0 · Basemap: Natural Earth. Analysis: A. Bouznad, CIRANO / Polytechnique Montréal

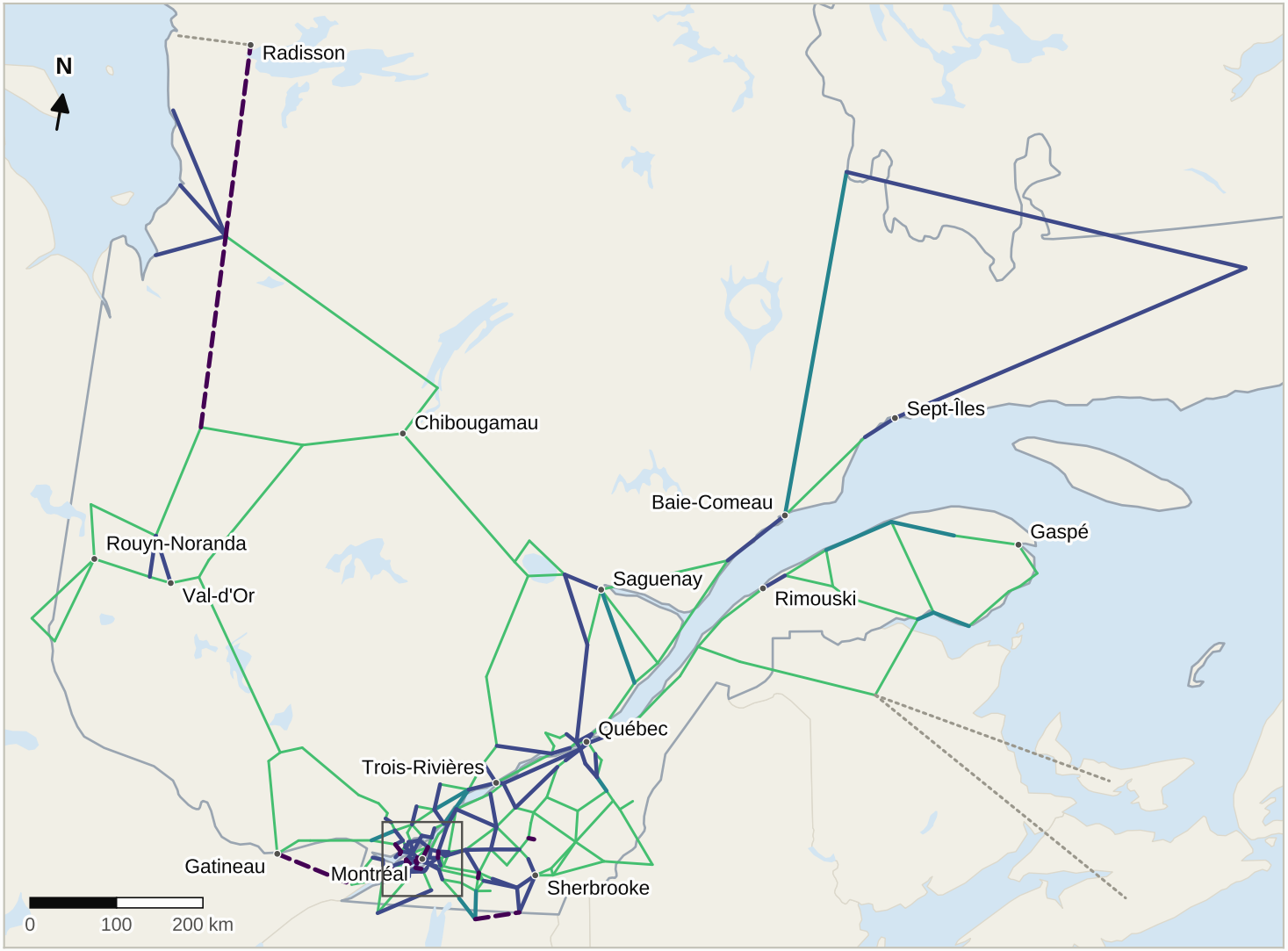
Question Which links carry the most traffic, and how unevenly is it spread?

Data & method 2025 AADT from MTMD counts, matched to each link's road segments (median 8 per link); link value = 80th-percentile segment (nearest rank). Vehicle-km = AADT × routed length in Quebec. Truck share is in the interactive map: only 72 of 285 links have a 2025 MTMD truck count.

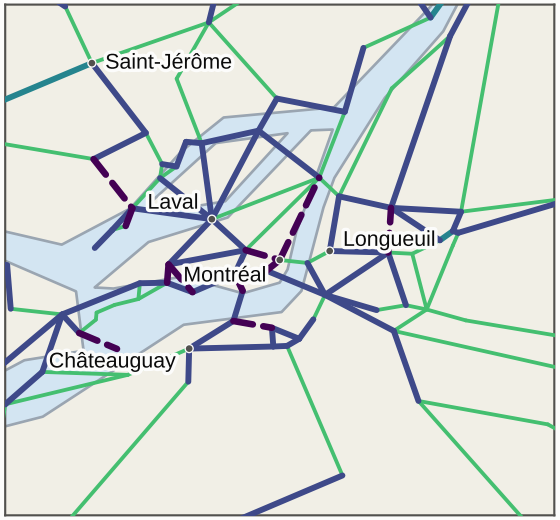
What it helps decide Ranks links for closure scenarios: losing a top-class link disrupts far more daily trips than a bottom-class one (median AADT 33,000 in the corridor vs 6,500 elsewhere).

166 of 307 links rest on a 2025 MTMD traffic estimate; 119 rely on imputation

Source of the AADT value used for each link, traced back to the MTMD segment that sets it (the link's 80th-percentile segment)



Zoom: Greater Montréal



Source of the link's AADT (links)

- 2025 MTMD estimate (166)
- Filled from earlier MTMD years (13)
- Imputed from nearby segments (k-NN) (106)
- Borrowed from the adjacent link (19)
- No MTMD data (3)

Data: MTMD traffic counts (AADT), Données Québec, CC BY 4.0 · Basemap: Natural Earth. Analysis: A. Bouznad, CIRANO / Polytechnique Montréal

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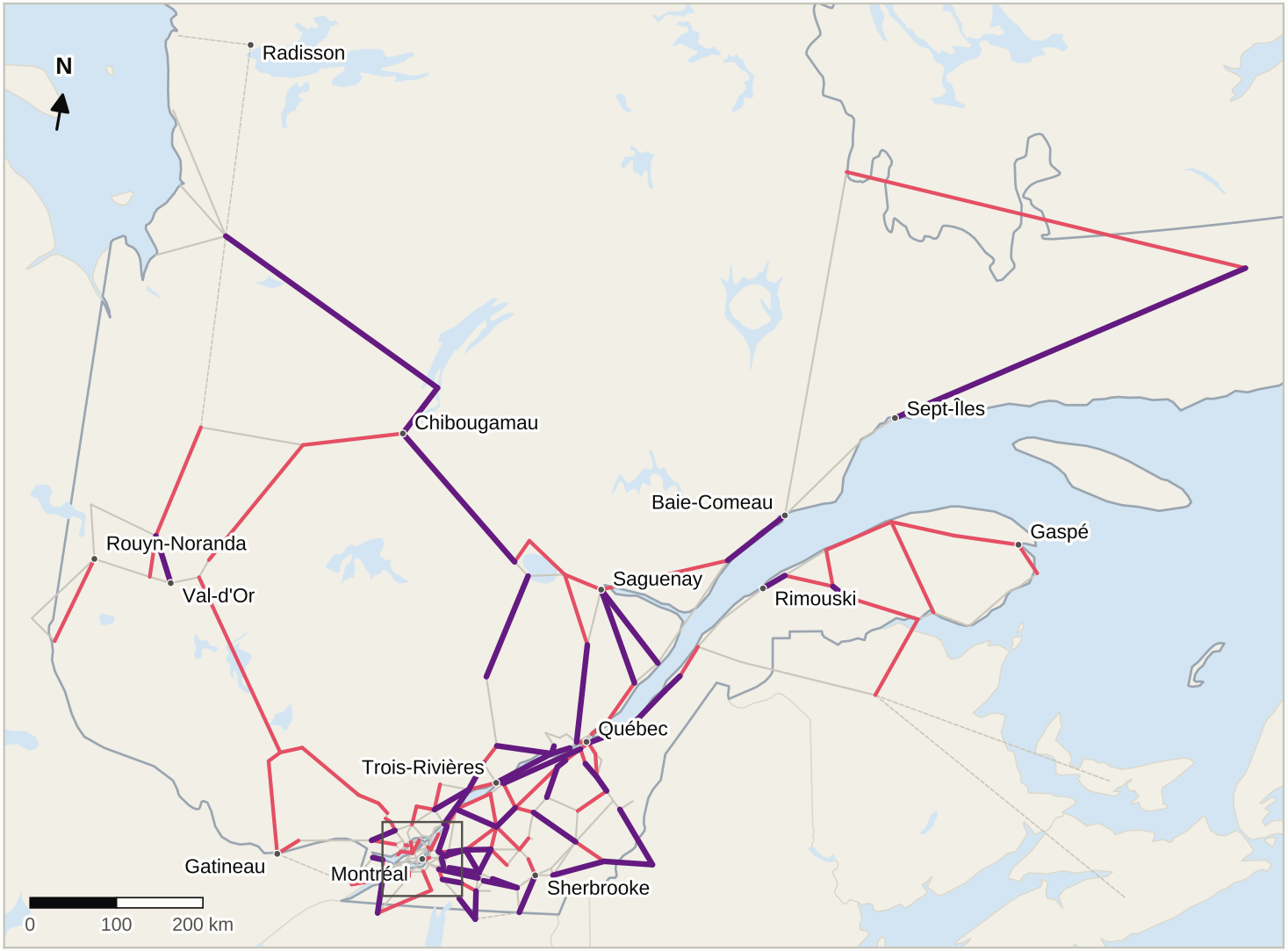
Question How much of each link's traffic value is observed, and how much is filled in?

Data & method For each link, the segment that sets its AADT is traced back to the MTMD file: a published 2025 value, a value filled from the segment's other years, or a k-nearest-neighbour imputation; links with no matched segment borrow the value of the adjacent link.

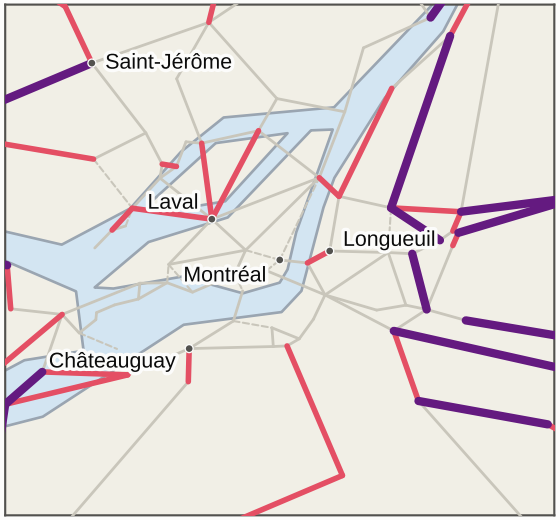
What it helps decide Shows where new traffic counts would most raise confidence before an investment or closure decision rests on a specific link.

134 of 285 links shift by more than 30% depending on the aggregation method

Spread of link AADT across 4 aggregation rules, (max − min) / mean, % · rules: mean, length-weighted mean, road-class-weighted mean, 80th percentile



Zoom: Greater Montréal



Spread across the 4 methods (links)

- > 60% (53)
- 30 – 60% (81)
- ≤ 30% (151)
- Not compared: borrowed or no MTMD data (22)

Data: MTMD traffic counts (AADT), Données Québec, CC BY 4.0 · Basemap: Natural Earth. Analysis: A. Bouznad, CIRANO / Polytechnique Montréal

Question	Does the result depend on how segment counts are aggregated into one value per link?
Data & method	Four aggregation rules (mean, length-weighted mean, road-class-weighted mean, 80th percentile); spread = (max − min) / mean. Links over 30% have a median of 11 matched segments, against 6 for the others.
What it helps decide	Flags the links where the aggregation choice must be justified, or a value range reported instead of a single figure.

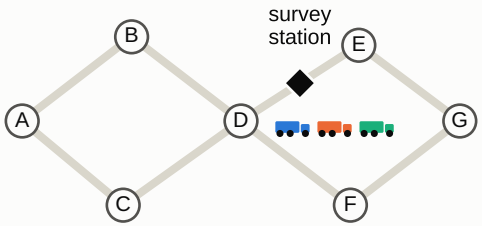
Phase 2 preview: turning a roadside truck survey into a dollar value per link

Method schematic on an illustrative network (no survey data shown) · trucks, commodities and paths are examples

PRELIMINARY – WORK IN PROGRESS

1 Survey

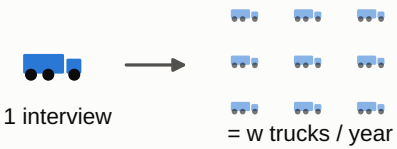
Trucks are interviewed at roadside stations: origin, destination, commodity, tonnes.



- A → G · lumber
- C → E · food
- B → E · empty

2 Value and weight

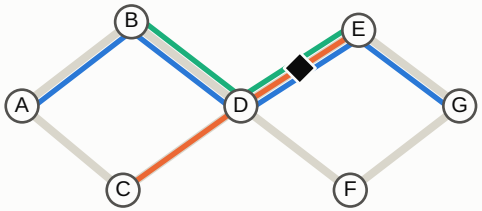
Each truck's cargo is valued in 2026 $; empty trucks count as $0. Each interview stands for w trucks a year.



- tonnes × $/t (lumber) × w_1
- tonnes × $/t (food) × w_2
- empty → $0 × w_3

3 Route

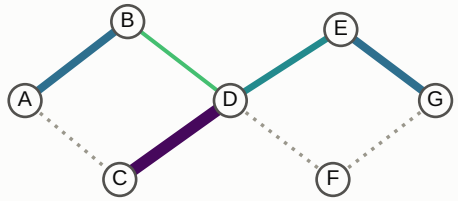
Each truck is routed from origin to destination on the graph and leaves its value on every link it uses.



- A → B → D → E → G
- C → D → E
- B → D → E

4 Aggregate per link

Weighted mean value per truck on each link, then × truck AADT × 365.
Links no surveyed truck uses are flagged.



- higher $ per truck
- lower $ per truck
- no surveyed truck: flagged

$$V_\ell = \frac{\sum_{i \in \ell} w_i \, t_i \, p_{c(i)}}{\sum_{i \in \ell} w_i} \times \mathrm{AADT}_\ell^{\mathrm{truck}} \times 365$$

V annual value of goods on link ℓ ($/year)
i surveyed trucks routed through ℓ
w survey expansion weight (trucks/year)
t tonnes carried (0 if empty)
p price per tonne ($/t) for commodity c, Statistics Canada CFAF, in 2026 dollars

Method: A. Bouznad, CIRANO / Polytechnique Montréal · Planned inputs: 2006-07 National Roadside Study (MTQ / Transport Canada), Statistics Canada CFAF (table 23-10-0142-01), Phase 1 truck AADT

Question	How can a roadside truck survey be turned into a dollar value of goods on each link?
Data & method	2006-07 National Roadside Study for origin, destination, commodity and weight; Statistics Canada CFAF for $ per tonne; truck AADT from Phase 1. No results shown: survey microdata are held under a data-sharing agreement and estimates are still in progress.
What it helps decide	Once complete, gives each link an economic weight, so closures can be ranked by the value of goods disrupted and not only by vehicle counts.